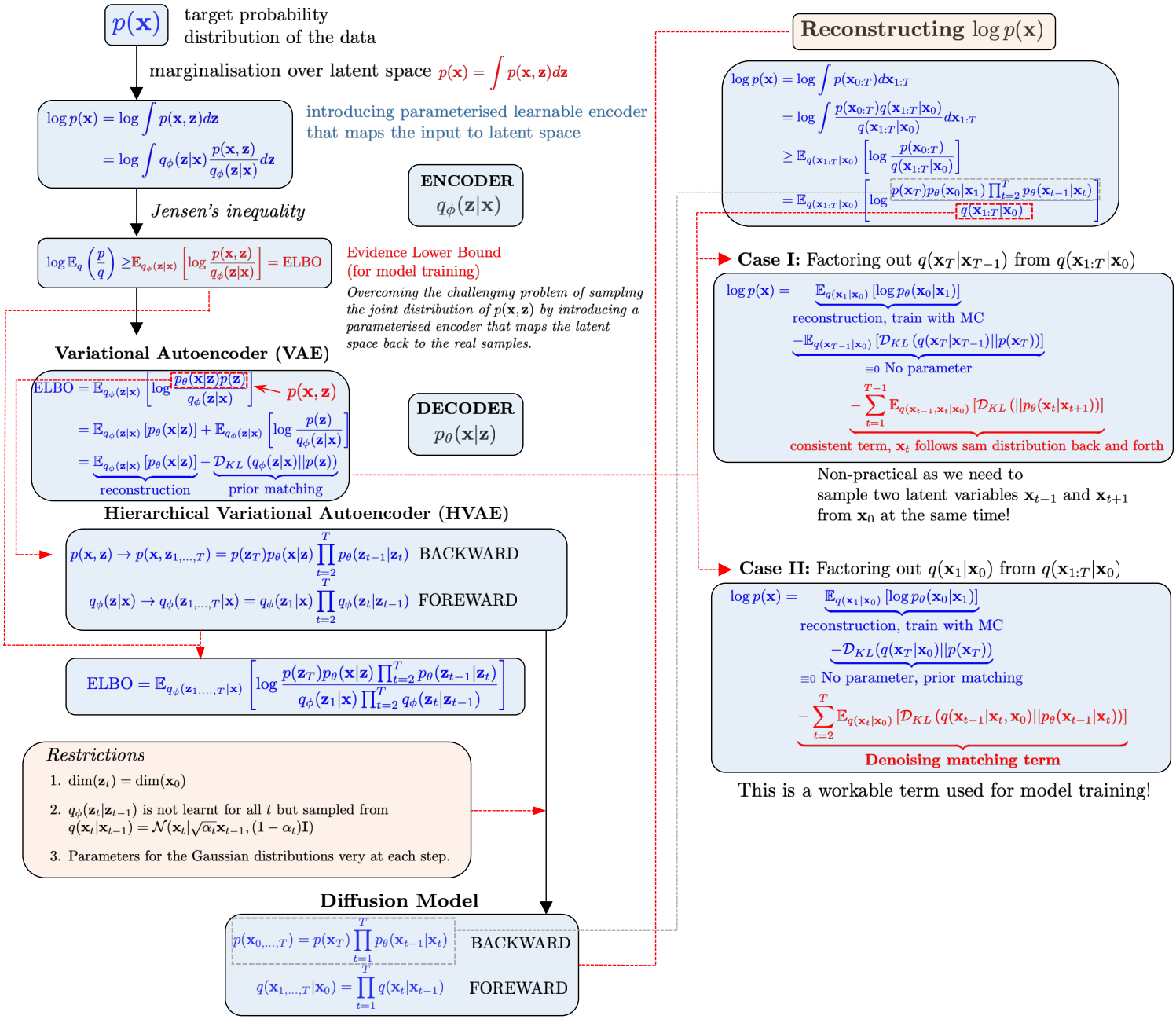

$p(\mathbf{x})$
target probability distribution of the data
marginalisation over latent space $p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z}$
$\log p(\mathbf{x}) = \log \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z}$
$= \log \int q_\phi(\mathbf{z}|\mathbf{x}) \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} d\mathbf{z}$
introducing parameterised learnable encoder that maps the input to latent space
ENCODER
$q_\phi(\mathbf{z}|\mathbf{x})$
Jensen's inequality
$\log \mathbb{E}_q \left(\frac{p}{q} \right) \geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] = \text{ELBO}$
Evidence Lower Bound (for model training)
Overcoming the challenging problem of sampling the joint distribution of $p(\mathbf{x}, \mathbf{z})$ by introducing a parameterised encoder that maps the latent space back to the real samples.
Variational Autoencoder (VAE)
$\text{ELBO} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right]$
$p(\mathbf{x}, \mathbf{z})$
$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [p_\theta(\mathbf{x}|\mathbf{z})] + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right]$
$= \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathcal{D}_{KL} (q_\phi(\mathbf{z}|\mathbf{x}) || p(\mathbf{z}))}_{\text{prior matching}}$
DECODER
$p_\theta(\mathbf{x}|\mathbf{z})$
Hierarchical Variational Autoencoder (HVAE)
$p(\mathbf{x}, \mathbf{z}) \to p(\mathbf{x}, \mathbf{z}_{1,\dots,T}) = p(\mathbf{z}_T) p_\theta(\mathbf{x}|\mathbf{z}) \prod_{t=2}^{T} p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t)$ BACKWARD
$q_\phi(\mathbf{z}|\mathbf{x}) \to q_\phi(\mathbf{z}_{1,\dots,T}|\mathbf{x}) = q_\phi(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q_\phi(\mathbf{z}_t|\mathbf{z}_{t-1})$ FOREWARD
$\text{ELBO} = \mathbb{E}_{q_\phi(\mathbf{z}_{1,\dots,T}|\mathbf{x})} \left[\log \frac{p(\mathbf{z}_T) p_\theta(\mathbf{x}|\mathbf{z}) \prod_{t=2}^{T} p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t)}{q_\phi(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q_\phi(\mathbf{z}_t|\mathbf{z}_{t-1})} \right]$
Restrictions
1. $\dim(\mathbf{z}_t) = \dim(\mathbf{x}_0)$
2. $q_\phi(\mathbf{z}_t|\mathbf{z}_{t-1})$ is not learnt for all t but sampled from $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$
3. Parameters for the Gaussian distributions very at each step.
Diffusion Model
$p(\mathbf{x}_{0,\dots,T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ BACKWARD
$q(\mathbf{x}_{1,\dots,T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})$ FOREWARD
Reconstructing $\log p(\mathbf{x})$
$\log p(\mathbf{x}) = \log \int p(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}$
$= \log \int \frac{p(\mathbf{x}_{0:T}) q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T}$
$\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$
$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_\theta(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$
Case I: Factoring out $q(\mathbf{x}_T|\mathbf{x}_{T-1})$ from $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$
$\log p(\mathbf{x}) =$
$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction, train with MC}}$
$\underbrace{-\mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} [\mathcal{D}_{KL} (q(\mathbf{x}_T|\mathbf{x}_{T-1}) || p(\mathbf{x}_T))]}_{\equiv 0 \text{ No parameter}}$
$\underbrace{-\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_t|\mathbf{x}_0)} [\mathcal{D}_{KL} (||p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1}))]}_{\text{consistent term, } \mathbf{x}_t \text{ follows sam distribution back and forth}}$
Non-practical as we need to sample two latent variables $\mathbf{x}_{t-1}$ and $\mathbf{x}_{t+1}$ from $\mathbf{x}_0$ at the same time!
Case II: Factoring out $q(\mathbf{x}_1|\mathbf{x}_0)$ from $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$
$\log p(\mathbf{x}) =$
$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{reconstruction, train with MC}}$
$\underbrace{-\mathcal{D}_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))}_{\equiv 0 \text{ No parameter, prior matching}}$
$\underbrace{-\sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\mathcal{D}_{KL} (q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{Denoising matching term}}$
This is a workable term used for model training!